CSCI 3203: Machine Learning I Syllabus

Fall 2026

Oklahoma City University, Petree College of Arts & Sciences, Computer Science

1. Instructor Information and Course Material

Professor:	Dr. Tashfeen, Ahmad (MSB 219B, `tashfeen@okcu.edu`[1]).
Office Hours:	Mondays, Wednesdays during 10–11 am and 5–6:30 pm or by appointment.[2]
Textbook:	Mitchell, Tom. *Machine Learning*. McGraw Hill, 1997.
Time & Location:	Monday, Wednesday at 11:00–12:15 pm in MSB 201.

2. Tentative Schedule

Week	Friday	Topics	Event	(%)
1	Aug 28	Maths & Python Refresher	Homework 1	5
2	Sep 4	Numpy, Matplotlib & Metrics	Homework 2	5
3	Sep 11	Naïve Bayes		
4	Sep 18	Naïve Bayes	Homework 3	10
5	Sep 25	Linear Regression		
6	Oct 2	Linear Regression	Homework 4	10
7	Oct 9	Review		
8	Oct 16	Review	Midterm	20
9	Oct 23		Fall Break	
10	Oct 30	Ridge Regression		
11	Nov 6	Ridge Regression	Homework 5	10
12	Nov 13	K-means		
13	Nov 20	K-means	Homework 6	10
14	Nov 27	Artificial Neural Networks	Thanksgiving	
15	Dec 4	Artificial Neural Networks		
16	Dec 11	Review	Homework 7	10
17	Dec 18	Review	Final	20

Table 1 — This may change depending upon class progress.

3. Grading Criteria

When accepted, work that is $n < 3$ days late will receive no more than $100(1 - 0.1n)\%$ of credit (10% deduction per day). For certain work, late submissions will not be accepted. Any submission that is not easy to read will receive a zero.

Total t %	Grade	Total t %	Grade	Total t %	Grade
		$t \geq 93$	A	$93 > t \geq 90$	A^-
$90 > t \geq 85$	B^+	$85 > t \geq 82$	B	$82 > t \geq 80$	B^-
$80 > t \geq 75$	C^+	$75 > t \geq 72$	C	$72 > t \geq 70$	C^-
$70 > t \geq 65$	D^+	$65 > t \geq 62$	D	$62 > t \geq 60$	D^-
$60 > t$	F				

Table 2 — A grade of Incomplete ("I") will only be assigned in case of documented extenuating circumstances. The "I" will be removed in accordance with university policy stated in the online undergraduate catalogue. Please also see subsection 4.4.

[1] Prefix [XXXX1234] followed by a space to all emails sent to the instructor, e. g., "[CSCI2114] Help with Homework 7."

[2] Zoom for a virtual meeting.

4. Policies & Resources

4.1. General Academic Guidelines. For an up-to-date version of the guidelines please visit the university sharepoint, see the OCU course schedule website for the finals week schedule and visit the online classroom.

4.2. Course Description. This concise, rigorously structured module introduces the core concepts of classification, regression and their associated evaluation metrics across supervised and unsupervised paradigms. Students will explore the underlying mathematical models, probability theory, loss functions and optimisation, before translating theory into practice with pure `NumPy` and `Matplotlib` implementations. Emphasis is placed on hands-on coding that reinforces algorithmic intuition, while visualisations illuminate model behaviour and performance. No external machine-learning libraries are used, ensuring a deep, from-scratch understanding of the fundamental techniques that underpin modern data-driven machine learning.

4.3. Course Learning Outcomes.

- Explain the fundamental concepts of supervised and unsupervised learning, including classification, regression and their principal evaluation metrics.
- Derive the mathematical foundations underlying common predictive and clustering models, and interpret associated loss functions and optimisation procedures.
- Implement these models from first principles using only `NumPy` and visualise results with `Matplotlib`.
- Evaluate model performance with appropriate quantitative metrics and visual diagnostics, and communicate the findings clearly.

4.4. Academic Integrity. The cheating rule for this class is simple: *don't turn in anything you did not understand*. I encourage you to get help and exhaust your resources. Though, if you turn in something (or answer a question with something you do not understand; can not explain) and I unsuccessfully ask you to explain your work, that will result in an *automatic F in the course* and disciplinary action will be taken. If you've been asked to demonstrate your work to the professor then you'll need to do so in his office hours or make an appointment. If you've been asked to demonstrate your work and you fail to do so, you will receive a zero in the assignment. For more, read the *Academic Honesty* section of the courses' catalogue.

4.4.1. *Artificial Intelligence (AI).* Help obtained by an AI, e. g., Large Language Model chat agent should be documented and turned in with the work. You may use AI to understand a general concept but it should *never* write or solve anything for you. A student who abuses AI will forfeit the right to use it for the remainder of the class.

4.5. Religious Accommodation. Oklahoma City University seeks to be supportive of religious observance among the members of our diverse campus community and to be as accommodating as possible. Students should discuss with their instructor at the beginning of the semester forms of religious observance (dress, fasting, specific prayer times) that may affect their full participation in the course. Students should also compare the class schedule to their own religious calendar to determine if there will be any class days in which the student expects to be absent due to the observance of a religious holiday. *Students must notify the instructor, in writing, of the expected absence within the first two weeks of the semester.* The instructor will then work with the student to develop a plan to reschedule any exams, assignments, or course activities for that day. The instructor, at his/her own discretion, will make reasonable accommodations wherever possible. Students should recognize that there may be some course aspects that cannot be rescheduled or accommodated, and it will therefore rest upon the student to determine whether they wish to remain enrolled in the course or have their grade potentially affected.

4.6. Mission Statement. The Petree College of Arts and Sciences provides a supportive, student-centered learning environment. Through an interdisciplinary curriculum that promotes active learning and individualized instruction, students are encouraged to think broadly, to find their passion, to collaborate with others, and to connect with the broader community.

4.7. Grook.

Put up in a place
where it's easy to see
the cryptic admonishment
T. T. T.

When you feel how depressingly
slowly you climb,
it's well to remember that
Things Take Time

——*Piet Hein (1970)*